

13. (2.25 points)

Given the following information for a reinsurer as of December 31, 2018:

Accident Year	Earned Risk Pure Premium (\$000)	Adjusted Premium (\$000)	Aggregate Reported Loss (\$000)	Aggregate Loss Report Lag
2015	6,800	7,400	4,500	90%
2016	7,200	7,700	3,400	70%
2017	7,600	7,900	2,300	50%
2018	8,000	8,000	3,000	30%

- Report lag reaches 100% in the 5th year.

a. (1 point)

Calculate the IBNR estimate for accident year 2018 as of December 31, 2018 using the Stanard-Bühlmann (Cape Cod) method.

b. (0.25 point)

Calculate the IBNR estimate for accident year 2018 as of December 31, 2018 using the chain ladder method.

c. (0.5 point)

Calculate the IBNR estimate for accident year 2018 as of December 31, 2018 using a credibility-weighted combination of the Stanard-Bühlmann and chain ladder methods given a selected credibility factor of 0.6.

d. (0.5 point)

Describe a situation in which an actuary would choose to use a credibility-weighted combination of the Stanard-Bühlmann and chain ladder methods.

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QUESTION 13

TOTAL POINT VALUE: 2.25

LEARNING OBJECTIVE(S): A13

SAMPLE ANSWERS

Part a: 1 point

Sample 1

$$ELR = (4500 + 3400 + 2300 + 3000) / (7400 \times 90\% + 7700 \times 70\% + 7900 \times 50\% + 8000 \times 30\%) = 0.717$$

$$8000 \times 0.717 \times (1 - 0.3) = 4015.2$$

Sample 2

Accident Year	Used Up Premium	Loss
2015	$7400 \times 0.9 = 6660$	4500
2016	5390	3400
2017	3950	2300
2018	2400	3000
Total	18400	13200

$ELR = 13200 / 18400 = 71.7\%$
 $AY\ 2018\ IBNR = (8000 - 2400) \times 71.7\% = 4015$

Part b: 0.25 point

Sample 1

$$R^{CL} = 3000 \times (1/0.3 - 1) = 7000$$

Sample 2

$$LDF = 1/30\% = 3.33$$

$$3000 \times 3.33 - 3000 = 7000\ (000)$$

Part c: 0.5 point

Sample 1

$$Z = 0.6 \times 30\% = 18\%$$

$$R = 18\% \times 7000 + (1 - 18\%) \times 4015 = 4552$$

Sample 2

$$Ult^{2018} = Lag \times Credibility \times (CL\ Ult) + (1 - Lag \times Credibility) \times (SB\ Ult)$$

$$= 0.3 \times 0.6 \times 10000 + (1 - 0.18) \times 7017.3973 = 7554.96$$

$$IBNR = 7554.96 - 3000 = 4554.96$$

Part d: 0.5 point

Sample Responses

- When the actuary wants to give more weight to the CL in older periods because they have more confidence in the loss data and less confidence in the on-leveled premiums used. Also want to give more weight to the SB method in the most recent period where they have more confidence in the on-level premium estimate and less confidence in the loss data.
- When the actuary doesn't have full faith in the chain ladder method b/c of highly leveraged LDFs but also doesn't have full faith in the on-level adjustments needed for the Cape Cod calculation.

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• If losses for the year are coming in higher than expected → CL method will more quickly reflect this so give some weight to CL and rest of weight to SB to not overreact and maintain stability. • For a long tail line where the chain ladder factors are highly leveraged, like workers comp. This would help moderate the early accident years where the LDFs are high but also respond to the emerging losses by gradually moving toward the CL as the ages mature.
EXAMINER'S REPORT
Candidates were expected to estimate IBNR estimates using the Cape Cod method, the chain ladder method, and the credibility-weighted combination of the Cape Cod and chain ladder methods. Candidates were also expected to describe a situation where the actuary would use the credibility-weighted method instead of only the Cape Cod method or the chain ladder method.
Part a
Candidates were expected to estimate the IBNR for accident year 2018 using the Cape Cod method. Common mistakes included: • Using the earned risk pure premium instead of the adjusted risk pure premium in the ELR calculation. • Subtracting the reported losses from the Cape Cod ultimate estimate instead of multiplying by 1 minus the report lag. • Calculating the ultimate loss estimate for accident year 2018 instead of the IBNR. • Calculating the IBNR estimate for all accident years instead of accident year 2018.
Part b
Candidates were expected to estimate the IBNR for accident year 2018 using the chain ladder method. Common mistakes included: • Calculating the ultimate loss estimate for accident year 2018 instead of the IBNR. • Calculating the IBNR estimate for all accident years instead of accident year 2018.
Part c
Candidates were expected to estimate the IBNR for accident year 2018 using the credibility-weighted combination of the Cape Cod and chain ladder methods. Common mistakes included: • Forgetting to multiply the credibility factor by the report lag. • Applying the credibility factor to the Cape Cod estimate and the remaining weight to the chain ladder method instead of the opposite. • Multiplying the credibility factor by the ELR instead of the report lag. • Calculating the ultimate loss estimate for accident year 2018 instead of the IBNR. • Calculating the IBNR estimate for all accident years instead of accident year 2018.
Part d
Candidates were expected to describe a situation where the actuary would use the credibility-weighted combination of the Cape Cod and chain ladder methods.

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Common mistakes included:

- Identifying a situation without providing any description as to why the credibility-weighted method is appropriate.
- Discussing credibility-weighting in general without describing the relative advantages and disadvantages of the Cape Cod and chain ladder methods.
- Briefly describing a situation in which the credibility-weighted method should be used without providing enough detail.